

S2 Supplementary Data

Complete contrast tables for MNPS, MNJ, regional/block Jacobian, EAP coupling, and physiological confounds

Article type: Supplement
Short title: S2 Data — MNJ Task Geometry (ds004511)

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S2.0 Complete global MNPS contrast table

All nine pre-specified pairwise contrasts (3 MNPS coordinates $\times$ 3 task pairs), Wilcoxon signed-rank test on session medians, BH-FDR applied across all nine simultaneously. Group medians (median of subject-level session medians) are shown per task. None survive correction — the global MNPS null reported as R2 in the main text.

Coord.	Contrast	GG med.	CC med.	Rest med.	<i>d</i>	FDR <i>q</i>	Sig.
m	GG vs CC	0.0237	0.0185	—	0.121	0.541	No
m	GG vs Rest	0.0237	—	0.0178	0.207	0.540	No
m	CC vs Rest	—	0.0185	0.0178	0.101	0.541	No
d	GG vs CC	0.0458	0.0473	—	-0.071	0.541	No
d	GG vs Rest	0.0458	—	0.0593	-0.269	0.541	No
d	CC vs Rest	—	0.0473	0.0593	-0.177	0.665	No
e	GG vs CC	0.0196	0.0285	—	-0.256	0.540	No
e	GG vs Rest	0.0196	—	0.0093	0.044	0.940	No
e	CC vs Rest	—	0.0285	0.0093	0.213	0.541	No

Table 1: Complete global MNPS pairwise contrast table (Wilcoxon signed-rank, session medians, BH-FDR over all 9 rows). $N = 44$ for GG-involving pairs, $N = 45$ for CC vs. Rest. Group medians are the median across subjects of each subject’s session-median coordinate value. No coordinate differentiates any task pair after correction. Friedman omnibus (3-condition): m $\chi^2 = 1.41$, $p = 0.49$; d $\chi^2 = 1.14$, $p = 0.57$; e $\chi^2 = 2.91$, $p = 0.23$. Source: 01_cross_task_mnps/mnps_pairwise.csv, mnps_cross_task_global.csv, mnps_friedman.csv.

S2.1 Complete MNJ contrast table

All 21 pre-specified pairwise contrasts (7 MNJ metrics $\times$ 3 task pairs), Wilcoxon signed-rank on session medians, BH-FDR across all 21 simultaneously. This is the source table for the summary shown in the main text (R3); note that the main-text statistical-design paragraph (§Methods) describes an 18-test primary family (6 metrics, excluding t trace) for the headline FDR claim, while this supplementary table reports the full 21-test family including t trace for completeness.

Metric	Contrast	GG med.	CC med.	Rest med.	d	FDR q	Sig.
frobenius_norm	GG vs CC	0.1560	0.1280	—	1.620	5.6e-12	Yes
frobenius_norm	GG vs Rest	0.1560	—	0.1310	0.738	2.8e-5	Yes
frobenius_norm	CC vs Rest	—	0.1280	0.1310	-0.229	0.284	No
rotation_norm	GG vs CC	0.0905	0.0798	—	1.005	7.7e-8	Yes
rotation_norm	GG vs Rest	0.0905	—	0.0861	0.098	0.331	No
rotation_norm	CC vs Rest	—	0.0798	0.0861	-0.401	0.034	Yes
trace	GG vs CC	-0.0035	0.0015	—	-0.240	0.182	No
trace	GG vs Rest	-0.0035	—	-0.0014	-0.138	0.350	No
trace	CC vs Rest	—	0.0015	-0.0014	0.086	0.671	No
spectral_radius	GG vs CC	0.0774	0.0624	—	1.808	5.6e-12	Yes
spectral_radius	GG vs Rest	0.0774	—	0.0599	1.979	5.6e-12	Yes
spectral_radius	CC vs Rest	—	0.0624	0.0599	0.268	0.147	No
rotational_power	GG vs CC	0.3699	0.4034	—	-1.025	1.1e-7	Yes
rotational_power	GG vs Rest	0.3699	—	0.4528	-1.410	8.1e-11	Yes
rotational_power	CC vs Rest	—	0.4034	0.4528	-0.710	7.8e-5	Yes
aci	GG vs CC	0.7662	0.8223	—	-1.015	1.0e-7	Yes
aci	GG vs Rest	0.7662	—	0.9097	-1.340	8.1e-11	Yes
aci	CC vs Rest	—	0.8223	0.9097	-0.713	6.3e-5	Yes
mdr	GG vs CC	0.7859	0.7497	—	0.420	0.006	Yes
mdr	GG vs Rest	0.7859	—	0.7078	1.022	3.1e-7	Yes
mdr	CC vs Rest	—	0.7497	0.7078	0.703	5.1e-5	Yes

Table 2: Complete MNJ pairwise contrast table (Wilcoxon signed-rank, session medians, BH-FDR over all 21 rows). $N = 44$ for GG-involving pairs, $N = 45$ for CC vs. Rest. 15/21 contrasts survive FDR 5%; all 6 non-significant contrasts involve trace, rotation_norm (GG vs. Rest), or frobenius_norm/spectral_radius (CC vs. Rest), consistent with the interpretation that CC and Rest resemble each other in absolute deformation magnitude while GG is the outlier condition. Source:

03_mnj_reachability/mnj_cross_task_wilcoxon.csv, subject_mnj_summary.csv.

S2.2 Regional MNPS contrasts (network-stratified)

MNPS coordinates were also tested within each of four cortical networks (frontal, central, parietal-occipital, temporal), applying BH-FDR separately within the 36-row family (4 networks $\times$ 3 coordinates $\times$ 3 task pairs). Full detail in mnps_network_pairwise.csv; the two rows surviving correction are shown below alongside the next-largest (uncorrected) effects for context.

Network	Coord.	Contrast	d	p	FDR q	Sig.
Central	m_median	GG vs CC	0.571	0.0008	0.020	Yes
Temporal	m_median	GG vs Rest	0.454	0.0011	0.020	Yes
Frontal	e_median	GG vs Rest	0.401	0.015	0.151	No
Temporal	e_median	GG vs CC	-0.336	0.019	0.151	No
Frontal	e_median	CC vs Rest	0.396	0.021	0.151	No
Frontal	d_median	GG vs Rest	-0.299	0.049	0.291	No

Table 3: Regional MNPS contrasts, top 6 of 36 rows by uncorrected p (Wilcoxon signed-rank, BH-FDR within the full 36-row network family). Only two rows survive correction: mobility (m) is higher in gambling than cognitive control in the central network, and higher in gambling than rest in the temporal network. Both are small, spatially localized departures from the global MNPS null (S2.0) rather than a broad regional effect — 34/36 rows are non-significant. All other network $\times$ coordinate

$\times$ contrast combinations had $q > 0.15$. Source: 01_cross_task_mnps/mnps_network_pairwise.csv.

The presence of two significant regional rows out of 36 (5.6%) is close to the false-positive rate expected under BH-FDR at $\alpha = 0.05$ applied to a family with a true null rate near 1; they should be interpreted as weak, localized signals rather than a confirmed regional MNPS effect, and are not treated as primary findings in the main text.

S2.3 Block-level Jacobian contrasts (independent aggregation pathway)

Block-level Jacobian summaries are computed from task-segment aggregation rather than epoch-level medians (S1.2), providing an independent cross-check of the epoch-level MNJ result.

Metric	Contrast	d	p	FDR q	Sig.
block_frobenius_mean	GG vs CC	1.695	1.1e-13	1.4e-12	Yes
block_frobenius_mean	GG vs Rest	0.596	1.4e-4	5.8e-4	Yes
block_frobenius_mean	CC vs Rest	-0.456	0.0098	0.015	Yes
block_trace_mean	GG vs CC	-0.258	0.0085	0.015	Yes
block_trace_mean	GG vs Rest	-0.030	0.986	0.986	No
block_trace_mean	CC vs Rest	0.176	0.105	0.125	No
block_anisotropy_mean	GG vs CC	0.388	0.0009	0.0028	Yes
block_anisotropy_mean	GG vs Rest	-0.146	0.029	0.038	Yes
block_anisotropy_mean	CC vs Rest	-0.151	0.615	0.671	No
c_rot_mean	GG vs CC	1.395	3.8e-12	2.3e-11	Yes
c_rot_mean	GG vs Rest	0.318	0.0025	0.006	Yes
c_rot_mean	CC vs Rest	-0.476	0.0047	0.009	Yes

Table 4: Block-level Jacobian contrasts ($N = 44/45$, Wilcoxon signed-rank, BH-FDR within the 12-row block family). Friedman omnibus: block_frobenius_mean $\chi^2 = 47.8$, $p < 10^{-10}$; c_rot_mean $\chi^2 = 34.4$, $p < 10^{-7}$; block_trace_mean $\chi^2 = 11.1$, $p = 0.004$; block_anisotropy_mean $\chi^2 = 7.8$, $p = 0.021$. The GG > CC Frobenius effect ($d = 1.70$) closely replicates the epoch-level result ($d = 1.62$, S2.1) using an entirely independent aggregation pathway. Source:

01_cross_task_mnps/block_jacobian_pairwise.csv, block_jacobian_friedman.csv.

S2.4 EAP-compatible respiratory-anchoring coupling (full table)

Session-level Spearman correlations between resp_anchor_index and MNJ metrics, computed separately within each task, with HRV (ecg_hrv_rmssd_ms) and cardiorespiratory RSA amplitude as additional physiological covariates for reference. BH-FDR applied within each task’s 6-metric family.

Task	MNJ metric	r (resp. anchor)	p	FDR q	Sig.
Gambling	frobenius_norm	-0.216	0.159	0.410	No
Gambling	rotation_norm	-0.219	0.153	0.410	No
Gambling	spectral_radius	-0.253	0.098	0.410	No
Gambling	rotational_power	-0.040	0.795	0.917	No
Gambling	aci	-0.040	0.795	0.917	No
Gambling	mdr	-0.058	0.707	0.917	No
Cognitive control	frobenius_norm	+0.020	0.895	0.917	No
Cognitive control	rotation_norm	+0.021	0.890	0.917	No
Cognitive control	spectral_radius	-0.058	0.703	0.917	No
Cognitive control	rotational_power	-0.016	0.917	0.917	No
Cognitive control	aci	-0.016	0.917	0.917	No
Cognitive control	mdr	+0.065	0.674	0.917	No
Rest	frobenius_norm	-0.157	0.303	0.681	No
Rest	rotation_norm	-0.226	0.135	0.410	No
Rest	spectral_radius	-0.062	0.687	0.917	No
Rest	rotational_power	-0.324	0.030	0.179	No†
Rest	aci	-0.324	0.030	0.179	No†
Rest	mdr	+0.352	0.018	0.179	No†

Table 5: Respiratory-anchoring $\times$ MNJ coupling by task (Spearman r , session medians, $N = 44\text{--}45$, BH-FDR within the 18-row full family used in the text). †: the rest-condition rotational_power/aci/mdr associations are the largest in the table (uncorrected $p < 0.05$) and are directionally consistent with the CC resp_anchor $\times$ m result reported in the main text (R6), but do not survive family-wise FDR ($q \approx 0.18$) and are reported as suggestive only. resp_anchor_index $\times$ m for cognitive control (the primary EAP-compatible finding, $r = -0.415$, $p = 0.0046$) is reported separately in S2.5 because it uses the

MNPS rather than MNJ metric family. Source: 03_mnj_reachability/mnj_resp_coupling.csv.

S2.5 EAP physio-MNPS coupling (global and per-network)

The 02_eap_physio_coupling pipeline additionally tested resp_anchor_index, ecg_hrv_rmssd_ms, and cardioresp_rsa_amplitude against all three global MNPS coordinates and the two block-Jacobian summaries, per task. The strongest cell in this larger family is the one highlighted in the main text:

Task	Physio.	Geometry	r	p	FDR q
Cognitive control	resp_anchor_index	m_median	-0.415	0.0046	0.206
Cognitive control	resp_anchor_index	d_median	+0.282	0.061	0.981
Gambling	resp_anchor_index	e_median	-0.211	0.168	0.981
Rest	cardioresp_rsa_amplitude	d_median	-0.185	0.223	0.981
Rest	resp_anchor_index	c_rot_mean	-0.181	0.233	0.981

Table 6: Top-5 physio-MNPS coupling cells by uncorrected p out of 45 tested (3 physiological measures $\times$ 5 geometry summaries $\times$ 3 tasks, BH-FDR across all 45). Only the cognitive-control resp_anchor_index $\times$ m cell approaches significance (uncorrected $p = 0.0046$) but does not survive the full 45-cell correction ($q = 0.206$); a targeted partial-correlation follow-up controlling for HRV (main text R6) was used to test this cell specifically rather than relying on the omnibus-corrected result. Full 45-row table: 02_eap_physio_coupling/coupling_per_task.csv, coupling_global.csv, coupling_per_network.csv.

S2.6 Physiological and temporal-structure confound partial-correlation table (combined)

Combined output of Script 05 (HR, respiration rate, respiratory anchor index, EOG blink rate), Script 07 (tonic EDA), Script 10 (corrugator EMG), and Script 11 (Sync-pulse event rate), testing whether GG–CC differences in each candidate confound track the GG–CC difference in MNJ Frobenius norm.

Confound (GG–CC difference)	r_s	p	FDR q	Sig.
Heart rate	-0.06	0.71	0.77	No
Respiration rate	-0.29	0.06	0.20	No
Respiratory anchor index	+0.33	0.03	0.20	No
EOG blink rate	-0.20	0.19	0.44	No
Tonic EDA (SCL)	-0.10	0.51	0.77	No
Corrugator EMG (RMS)	-0.08	0.60	0.77	No
Sync-pulse event rate	-0.05	0.77	0.77	No

Table 7: Physiological/temporal-structure confound audit for the primary GG>CC Frobenius-norm effect ($N = 44$ intersection sample, Spearman correlation of GG–CC difference series, BH-FDR over 7 rows; eog_artifact_fraction excluded for zero variance). No candidate confound survives correction. Source: 05_mnj_confound_audit/mnj_physio_confound_partial.csv, 07_eda_confound_control/full_confound_table.csv, 10_corrugator_emg_confound/full_confound_table_with_emg.csv, 11_event_density_sensitivity/full_confound_table_with_event_rate.csv.

S2.7 EDA descriptive summary by task

Session-level EDA extraction summary (Script 06), 129 of the 134 valid EEG sessions had usable EDA data (five sessions across three subjects failed the session-level EDA quality gate; see S1.3).

Task	Mean epochs/ session	Mean QC pass rate	Median tonic SCL (μ S)
Rest	126	99.7%	2.06
Cognitive control	262	98.3%	3.22
Gambling	910	98.9%	3.35

Figure 1: EDA extraction summary by task (55,903 epochs total, 129 sessions). Both active tasks (GG, CC) show significantly higher tonic SCL than Rest (GG vs. Rest: $d = 0.56$, $p = 0.0002$; CC vs. Rest: $d = 0.67$, $p < 0.0001$); GG and CC are statistically indistinguishable from each other ($d = -0.07$, $p = 0.55$). Source: 06_eda_extraction/eda_qc_summary.csv, eda_features.parquet.

Bibliography